

Convergence in distribution of the normalised remainders

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2026-09-07

Abstract

Completion of Astra #11 rank 4. For measurable random Taylor data $X_n : \Omega \rightarrow \mathbf{CoeffPair}$ with $\|X_n\| \leq M$, converging in distribution to Z , and a scale $N_n \rightarrow \infty$, the ordered normalised remainders of the chart integral converge in distribution to the limiting canonical coefficients:

$$\frac{Z_{N_n}(X_n) - L_{<\alpha}}{N_n^{-\alpha} \log N_n} \Rightarrow A_\alpha(Z), \quad \frac{Z_{N_n}(X_n) - L_{<\alpha} - N_n^{-\alpha} A_\alpha(X_n) \log N_n}{N_n^{-\alpha}} \Rightarrow B_\alpha(Z)$$

(`tendstoInDistribution_normA`, `tendstoInDistribution_normB`), for every retained exponent α of the Taylor tree. This is the $d = 2$ chart-level form, for bounded analytic families, of the second conclusion of `thm:strataempiricalexpansion`. Zero `sorry/axiom`.

1 The things formalised

```
theorem exists_gap (P : Finset ) ( ) : , 0 < P, < → +
def ballCoeffBound ( b M) (h h k k) (P : Finset ) :
  -- Σ_{P} (|envelope bound for A_| + |envelope bound for B_|)
theorem coeff_abs_le_ball ... (a) (ha : ||a|| M) ( ) (h : P) (h0 : 0 < ) :
  |coeffA ... a| ballCoeffBound ... P |coeffB ... a| ballCoeffBound ... P
def normA ( b p p T) (h h k k) ( N) (a : CoeffPair) : :=
  (chartZ ... N a - lowerPart (polesBelowGen ...)) (coeffA ... · a) (coeffB ... · a) N
  / (N^(-) * log N)
def normB ... :=
  (chartZ ... N a - lowerPart ... - N^(-) * (coeffA ... a * log N)) / N^(-)
theorem normA_sub_le / normB_sub_le ... (h : polesBelowGen ...) ( ) (hgap) (N)
  (hN : exp 1 N) :
  |normA ... N a - coeffA ... a|
    errA |uniformTreeConst b M (2M) p p T h k 0| (ballCoeffBound ...) (2T) P N
theorem measurable_lowerPart_comp / measurable_normA_comp / measurable_normB_comp
theorem tendstoInMeasure_of_eventually_abs_le ( ) (f : → Ω → ) (e : → )
  (he : Tendsto e l ( 0)) (hf : n in l, , |f n| e n) :
  TendstoInMeasure f l (fun _ => 0)
theorem tendstoInDistribution_normA ... [l.IsCountablyGenerated] (M) (hM : 0 M)
  (X : → Ω → CoeffPair) (hXm : n, Measurable (X n)) (hXb : n , ||X n || M)
  (Z : Ω' → CoeffPair) (hX : TendstoInDistribution X l Z (fun _ => ) ' )
  (Nseq : → ) (hN : Tendsto Nseq l atTop)
  ( ) (h : polesBelowGen h h k k p p T) :
  TendstoInDistribution (fun n => normA b p p T h h k k (Nseq n) (X n )) l
  (coeffA h h k k Z) (fun _ => ) '
theorem tendstoInDistribution_normB ... : ... normB ... coeffB b h h k k Z
```

2 Mathematics

Write the normalised remainder as $A_\alpha(X_n) + \varepsilon_n$. On the ball $\|a\| \leq M$ the uniform Taylor tree (unit 149) gives the remainder bound with the data-independent constant $K = |\text{uniformTreeConst}|$, the envelope bounds (unit 134) give a uniform coefficient bound K_c (a sum of absolute values of the per-exponent envelope constants, so that each term is dominated without a separate nonnegativity argument), and the exponent set has a gap δ above α ; so unit 148's deterministic bound gives $|\varepsilon_n(\omega)| \leq \text{errA}(N_n) \rightarrow 0$ uniformly in ω . Hence $\varepsilon_n \rightarrow 0$ in probability (indeed the exceptional set is eventually empty), $A_\alpha(X_n) \Rightarrow A_\alpha(Z)$ by the continuous mapping theorem (unit 147), and Slutsky's theorem gives the claim. The B-slot is identical with the current $A_\alpha(X_n)$ removed first, exactly as Astra #11 insisted.

3 Paper-facing meaning

thm:strataempiricaexpansion states that

$$\frac{Z_n^0 - \sum_{(\mu,m) < (\mu',m')} C_{\mu,m}(\xi_n) n^{-\mu} (\log n)^{m-1}}{n^{-\mu'} (\log n)^{m'-1}} \rightarrow C_{\mu',m'}(G)$$

in distribution. For $d = 2$, one chart, and Taylor data confined to a norm ball, this is now a theorem: the two slots $(\alpha, 2)$ (the log coefficient) and $(\alpha, 1)$ (the constant) are handled in order. Limitations, to be stated in staging: bounded families (the natural upgrade localises by $\Pr(\|X_n\| > M)$); a common sample space (Mathlib's Slutsky is stated for a fixed Ω); the functional CLT $\xi_n \Rightarrow G$ and the resolution/chart assembly are not formalised.

4 Lean notes

- `Finset.exists_min_image` on the filter $\{\gamma \in P : \alpha < \gamma\}$ gives the gap; the empty case takes $\delta = 1$.
- No nonnegativity lemmas exist for the envelope constants `canonUEnv` etc.; wrapping each per-exponent bound in `|\cdot|` and summing makes `Finset.single_le_sum` applicable with `abs_nonneg`. Likewise $K = |\text{uniformTreeConst}|$ avoids proving the constant nonnegative.
- `tendstoInMeasure_iff_norm` turns convergence in measure into statements about $\mu\{\varepsilon \leq \|f_n - g\|\}$; when $|f_n| \leq e_n < \varepsilon$ everywhere the set is empty and `tendsto_const_nhds.congr'` finishes. `Set.mem_setOf_eq` is deprecated for `Set.mem_ofPred_eq`.
- `TendstoInDistribution.add_of_tendstoInMeasure_const` needs `[1.IsCountablyGenerated]` and a fixed sample space; the final `congr` step is `simp only [Pi.add_apply, Function.comp]`; ring.